Mini-Project 4: Object Oriented Programming

Lab Report

Name: Jiawei Zhu

Student ID Number: 923312684

Instructions

Download this lab report template and complete each section. Be sure to include this cover page along with your report. Upload your plots by copy-pasting them into this document. Submit a ***pdf*** of your completed lab report to Canvas by the deadline.

You may submit as many times as you wish up until the deadline. However, late submissions are not accepted. Questions begin on the following page and there is one question per page. When uploading your submission, **do not change the order of questions or answer more than one question per page.** Doing so may cause your submission to be imported incorrectly.

Note: it is a violation of academic integrity to submit code written by someone else or plots produced by someone else or by using another person's code.

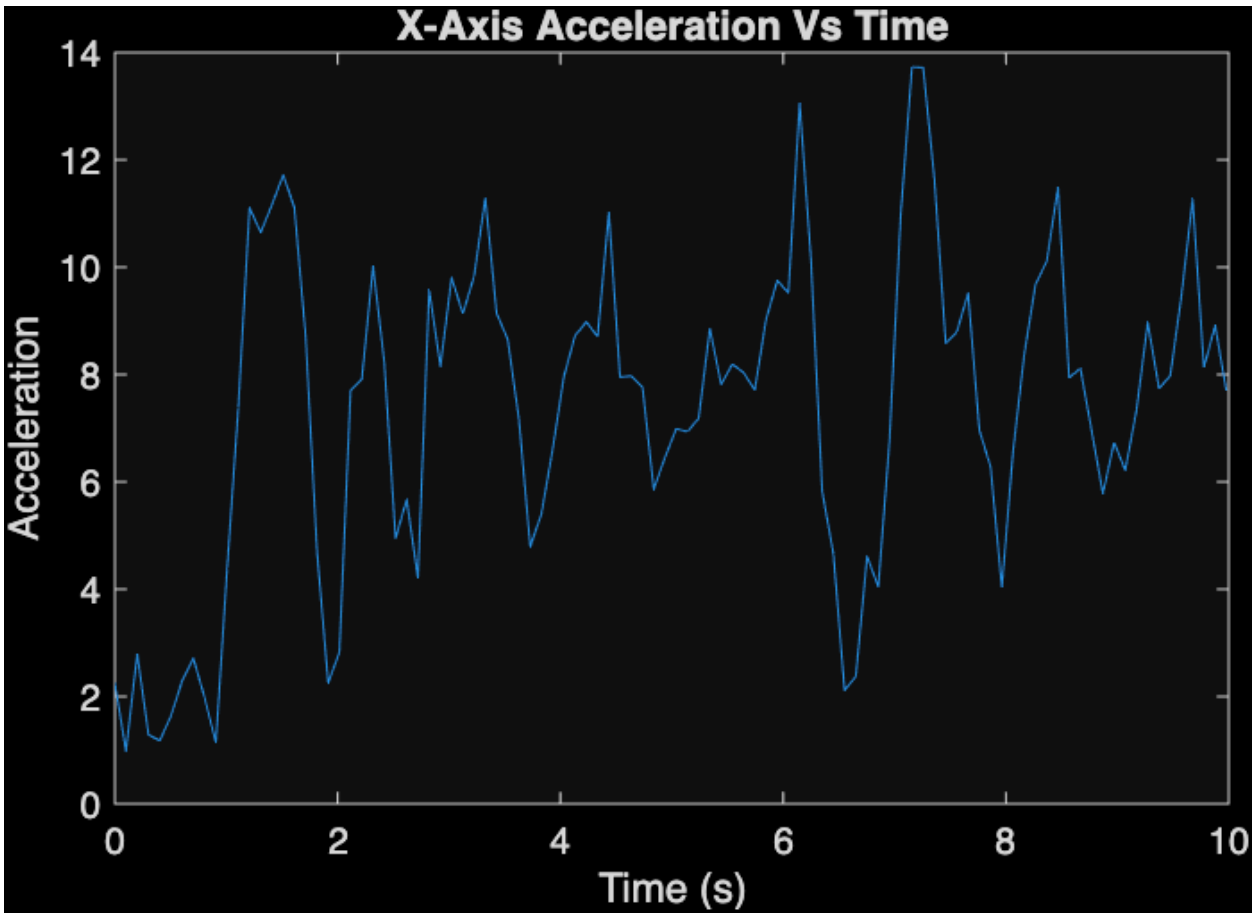
Question 1: Summary of the Project

In your own words, write a short summary of what was done for this project.

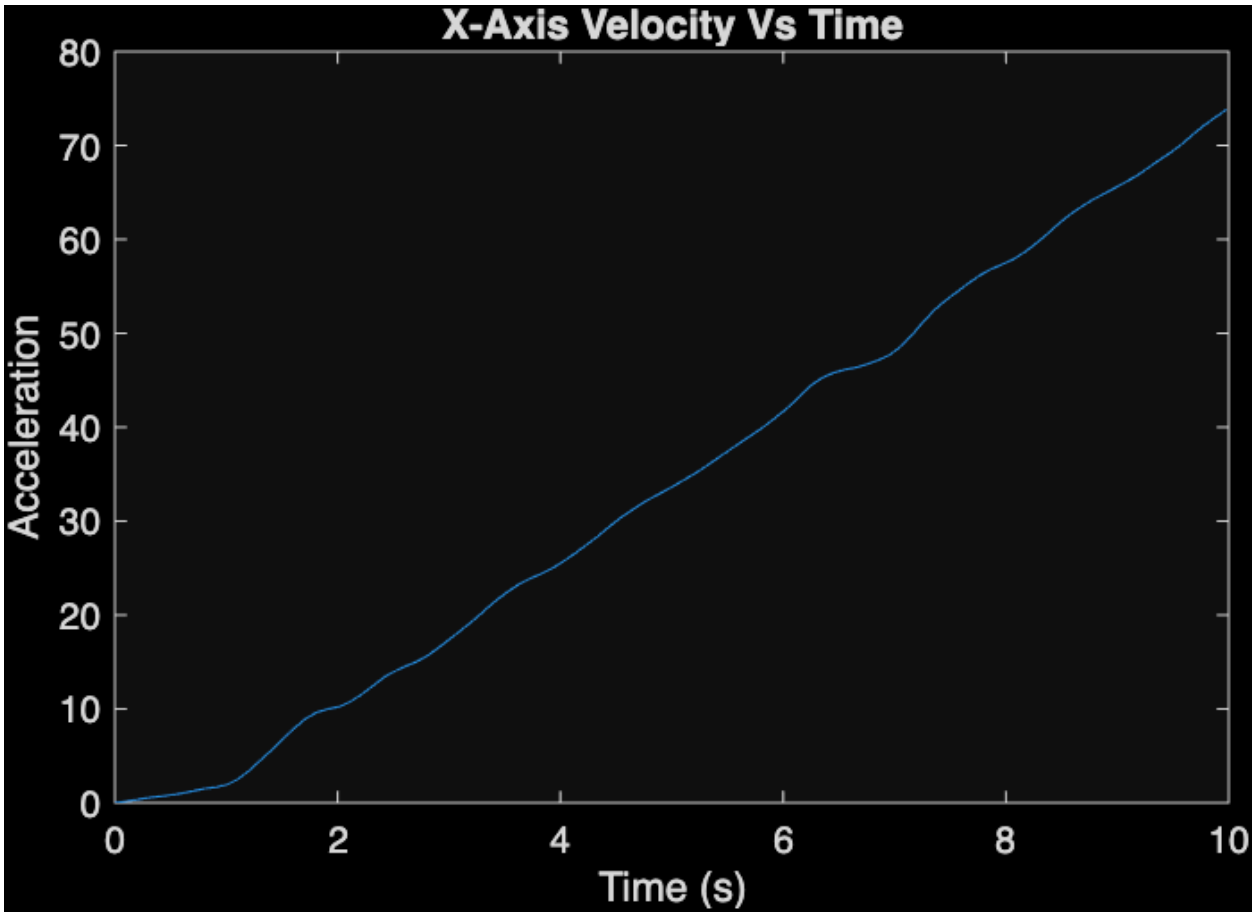
This mini project is using the SensorDataAnalysis.m file to complete data collection. First of all, using Mobiledev to connect the phone for recording for about 10 seconds while I was moving my phone. Then I used the class methods to choose the X-axis acceleration and applied basic calculus: integrate first to get its velocity, and do one more integration to get its displacement. Showing these graphs can show how the acceleration, velocity, and displacement change.

Question 2: Data Analysis

Part A) This question refers to your results from section 2.1 in the lab manual. Include your plot of acceleration of time (e.g. the raw results from the accelerometer)



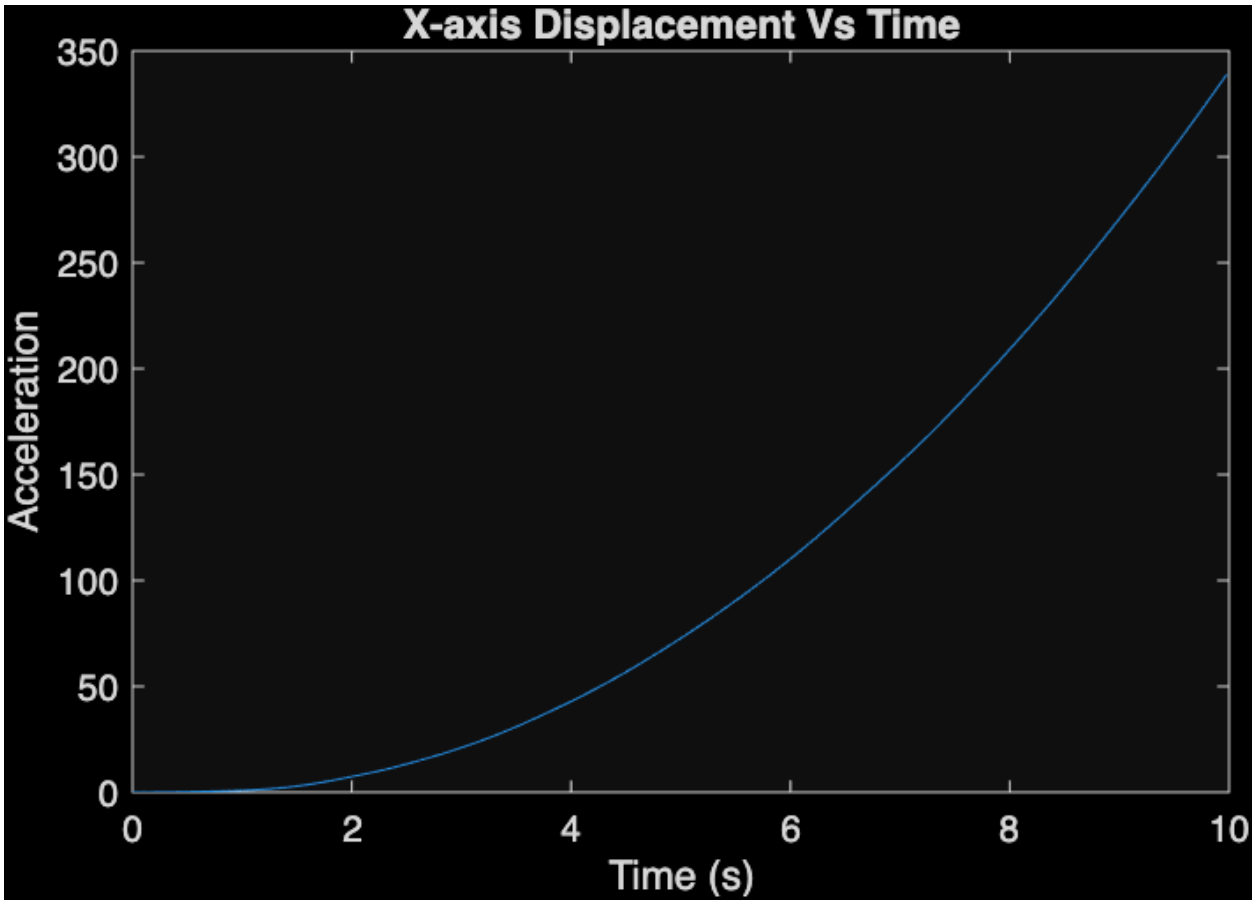
Part B) This question refers to your results from section 2.1 in the lab manual. Include your plot of velocity over time as determined by numerically integrating your accelerometer measurements.



Part C) This question refers to the plot from part (B). Does your plot match your intuition about velocity? Your response should be specific and quantitatively address data from your plot.

The plot doesn't match my intuition velocity. In the plot, it shows that I only use 10 seconds to add 75 units. But actually, I was not always increasing my velocity, but the plot does. During that time I was increase my speed first, but that time I stopped. I think this problem is because of some bias in the integration in such positive data. So this plot didn't show the real motivation.

Part D) This question refers to your results from section 2.1. Include your plot of displacement over time as determined by numerically integrating your *velocity* measurements.



Part E) This question refers to the plot from part (D). Does your plot match your intuition about position? Your response should be specific and quantitatively address data from your plot. Note that, if you return to exactly where you started, your net displacement is zero. Does your plot agree with this? If not, why not?

The plot doesn't match my intuition about position. The displacement increases from 0 to 340 units in 10 seconds. During that time, I went back to the point I started, but the displacement didn't go back to 0 and was still increasing. One of the reasons is that in part C, I said it is a wrong/biased integration. With this wrong data integration, it is one more time, the data can only be worse. And also, there is some gravitational and sensor noise in this environment, which can also influence the result. So this plot didn't show the real motivation.

Question 3: AI Reflection

If you used AI for any aspect of your lab, you must complete a short reflection documenting your AI use.

If you did not use AI, write your name after the following statement:

I did not use AI as part of this assignment, I understand it is a violation of academic integrity to use AI without disclosing it**: <your name here>**

You must also attach a log (either as a text log, or screenshots) of your interaction with AI. Your log does not need to be exhaustive, but should provide a reasonable overview of your interaction with the AI: you should include the “important” questions you asked but can omit the complete text of the AI’s response and minor follow up questions. An example is provided on Canvas. Provide 1-3 sentences in answer to the following questions:

1. **Which AI tool(s) did you use? E.g. cursor, ChatGPT, Gemini, etc…**
   ChatGPT

2. **Describe how you used AI in your lab. E.g. to help with debugging, to suggest ways to improve efficiency/clarity of your code, etc…**
   I used AI to help me debug in the test. When I run the coding assignment, all the test looks good except that getTargetSignal throws an error if no TargetSignal is set doesn’t work. I asked for some suggestions on how to fix this function.

3. **What steps did you take to verify that the responses given to you by the AI were good ones?**
   I checked the suggestion from ChatGPT for adding an if function to determine whether there is a TargetSignal or not. To check it works or not, I checked the error message: “Your getTargetSiqnal function does not throw an error with no set targt signal. But I think some of its suggestion is uncesary, so I changed to

```
 if isempty(obj.TargetSignal)
error('No TargetSignal is set.');
end
```

   After testing the code in MATLAB and MATLAB grader, all the tests pass, so that works. So I kept this suggestion.

Attach chat logs here

Hi, I'm an undergraduate student taking an engineering class. I have some problems with my mini project. I am not sure why this part doesn't work. Could you give me some suggestions for debugging this function? function [signal, Ts] = getTargetSignal(obj)

```
    % Extract the traget signal and time stamp from the properties
    signal = obj.TargetSignal;
    Ts = obj.TargetSignalTs;
    % and return it here.
end
```

Sure, here's a simple answer you can use:

You can fix this by adding a check at the start of the function and calling `error` if the properties are empty, for example:

matlab 复制代码

```matlab
function [signal, Ts] = getTargetSignal(obj)

    % Check if a target signal has been set
    if isempty(obj.TargetSignal) || isempty(obj.TargetSignalTs)
        error('No TargetSignal is set. Call setTargetSignal first.');
    end

    % Extract the target signal and time stamp from the properties
    signal = obj.TargetSignal;
    Ts     = obj.TargetSignalTs;
end
```

After you add this `if` statement, try creating a new `sensorDataAnalyzer` object and calling `getTargetSignal` **before** `setTargetSignal` to confirm that it throws an error, and then run the MATLAB Grader tests again.